



Measures of Location and Spread

Worksheet 1

Statistics Chapter 2: Measures of Location and Spread. Answer all questions. Show your working clearly. Total: 82 marks.
Time: 100 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are four sections, one for each topic in Chapter 2. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Write **estimate** whenever a mean or median comes from grouped data.
- Give non-exact answers to 3 significant figures unless stated otherwise.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Measures of central tendency	19	
B Measures of spread	19	
C Coding data	20	
D Outliers and skewness	19	
Roar-level challenge	5	
Total	82	

Section A Measures of central tendency

19 marks

1. Find the mean and the median of the data 4, 7, 9, 12, 15, 18. [2]

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2. The table shows the number of pets owned by 40 students.

[3]

Number of pets, x	0	1	2	3	4
Frequency, f	5	8	12	9	6

Find the mean, the median and the mode of the number of pets.

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3. The masses, m kg, of 40 parcels are shown in the table.

[3]

Mass, m (kg)	$0 \leq m < 10$	$10 \leq m < 20$	$20 \leq m < 30$	$30 \leq m < 40$	$40 \leq m < 50$
Frequency	4	11	15	7	3

Calculate an estimate of the mean mass of the parcels.

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4. Using the data in Question 3,

[4]

- (a) use linear interpolation to estimate the median mass,
- (b) explain why your answers to Question 3 and part (a) are only estimates.

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5. (a) The mean of 10 numbers is 12. When an eleventh number, x , is added, the mean becomes 13. [2]
Find x .

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- (b) The waiting times, t minutes, of 40 patients at a clinic are summarised below. [4]

Time, t (min)	$0 \leq t < 5$	$5 \leq t < 10$	$10 \leq t < 20$	$20 \leq t < 40$
Frequency	6	14	16	4

Estimate the mean and the median waiting time.

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- (c) Suggest why the median might be a better average than the mean for these data. [1]

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(Total for Question 5 is 7 marks)



Section B Measures of spread

19 marks

6. Find the range and the interquartile range of 3, 5, 6, 8, 9, 11, 14, 20.

[2]

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7. For a sample of 10 values, $\sum x = 120$ and $\sum x^2 = 1580$.

[3]

Find the mean and the standard deviation.

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8. The table shows the scores of 30 students on a quiz.

[3]

Score, x	1	2	3	4	5
Frequency, f	3	7	10	6	4

Find the mean and the standard deviation of the scores.

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9. Using the parcel data from Question 3, estimate the standard deviation of the masses.

[4]

10. Two groups of students sit the same test.

Group A: 12 students, mean score 20, standard deviation 3. Group B: 8 students, mean score 25, standard deviation 4.

(a) Find $\sum x$ and $\sum x^2$ for each group.

[3]

(b) Find the mean and the standard deviation of the scores of all 20 students together.

[4]

(Total for Question 10 is 7 marks)



Section C Coding data

20 marks

11. Data are coded using $y = x - 50$. The coded data have mean 3.2 and standard deviation 1.5. [2]
Find the mean and standard deviation of x .

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12. Data are coded using $y = \frac{x - 200}{10}$. The coded data have mean 4.5 and standard deviation 2.1. [3]
Find the mean and standard deviation of the original data.

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13. The masses, x grams, of 20 bags of flour are coded using $y = \frac{x - 1000}{50}$, giving $\sum y = 36$ and $\sum y^2 = 110$. [4]
Find the mean and the standard deviation of the masses.

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14. The temperatures in a city over a month have mean 15°C and standard deviation 4°C . [3]
Given that $F = 1.8C + 32$, find the mean and standard deviation of the temperatures in $^\circ\text{F}$.

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15. The prices, £ x , of 40 houses are coded using $y = \frac{x - 150\,000}{1000}$.

The coded data give $\sum y = 520$ and $\sum y^2 = 12\,000$.

(a) Find the mean and the standard deviation of y . [3]

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(b) Hence find the mean and the standard deviation of the house prices. [3]

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(c) A house priced at £250 000 is added to the data. Find the new mean house price. [2]

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(Total for Question 15 is 8 marks)

**Section D Outliers and skewness**

19 marks

16. A data set has $Q_1 = 20$ and $Q_3 = 32$. An outlier is any value more than $1.5 \times \text{IQR}$ beyond the quartiles. [2]
Is the value 55 an outlier? Show your working.

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17. A distribution has mean 45, median 40 and mode 36. [3]

- (a) Describe the skewness of the distribution, giving a reason.
- (b) State, with a reason, which average best represents the data.

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18. The data below show the numbers of emails received by 9 people in one day. [4]

2, 4, 5, 5, 6, 7, 8, 9, 25

Find the quartiles, and determine whether 25 is an outlier using the $1.5 \times \text{IQR}$ rule.

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19. For a data set the mean is 50 and the standard deviation is 6. An outlier is defined as any value more than 2 standard deviations from the mean. [3]
Determine whether the values 35 and 63 are outliers.

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20. The reaction times, in seconds, of a group of drivers have minimum 0.21, $Q_1 = 0.32$, median 0.36, $Q_3 = 0.45$ and maximum 0.82.

- (a) Show that 0.82 is an outlier, using the $1.5 \times \text{IQR}$ rule. [2]

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- (b) Describe the skewness of the distribution, giving a reason. [2]

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- (c) The mean is 0.39 s and the standard deviation is 0.12 s. Determine whether 0.82 is also an outlier using “more than 2 standard deviations from the mean”, and state, with a reason, which average and measure of spread are most suitable for these data. [3]

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(Total for Question 20 is 7 marks)



ROAR-LEVEL CHALLENGE

5 marks

21. For 10 values of x , $\sum(x - 20) = 15$ and $\sum(x - 20)^2 = 250$.

[5]

(a) Find the mean and standard deviation of x .

(b) An eleventh value, $x = 30$, is added. Find the new mean and standard deviation.

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END OF WORKSHEET

Total: 82 marks